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While moral dumbfounding has reinvigorated the debate surrounding emotion's involvement in moral judgement-making, its discovery has not provided evidence in favour of any specific emotions-based approach. Proponents of one theory, the Linguistic Analogy, argue that this approach has an advantage over moral intuitionism in its ability to accommodate contextual information, such as intentions and consequences, in the formation of a moral judgement. I argue that a dual-process intuitionism is capable of this level of complexity, in the same way that another family of intuitions — safety intuitions — are sensitive enough to be considered a source of information in safety scenarios.

Sensitive Intuitions

The introduction of moral dumbfounding has coincided with a great shift in how moral judgement-making is thought to occur. When an individual maintains a moral judgement, despite being unable to provide their reasoning behind it, they are said to be morally dumbfounded. This phenomenon presents strong evidence against the idea that morality is a fully rational process, and instead suggests that there is some emotive component that plays a fundamental role in forming moral judgements and advising moral action. However, the exact process by which emotions, or intuitions, or some other cognitive system advises morality is still hotly contested.

In this paper I argue that at its best, moral decision-making stems from automatic, rapid, subconscious processes and is advised and moulded by analytic thought. I do this by first walking through Jonathan Haidt and Craig Joseph's nativist intuitionism and Susan Dwyer's linguistic analogy as two potential candidates for an emotion-based conception of morality. I do my own extrapolation on Haidt and Joseph's intuitionism as best being viewed as dual-process intuitionism. Then, I address Dwyer's criticism that the intuitionist approach lacks the sensitivity and complexity that we may expect from the process responsible for forming moral judgements. I do so by making an analogy of my own between moral intuitions and safety intuitions, or "gut feelings." Lastly, I argue that the linguistic analogy may be at a disadvantage of its own, in its lack of ability to explain intra-cultural variance in responses to acceptability scenarios.

PART 1: Moral Dumbfounding and an Emotion-Based Morality

Although its precise definition is often debated, moral dumbfounding (MD) generally refers to the phenomenon whereby an agent maintains a moral judgement while knowingly providing an unsatisfactory explanation for it, or in some cases, being unable to provide any insight into the

thought process behind the judgement at all. Taboo scenarios are often utilized to invoke MD (however, MD is not necessarily only found in morally taboo situations). For example, Haidt et al. present a scenario where two siblings decide to have sex just to see what it's like (2000, p.15). They use contraception so there is no chance of pregnancy or spreading disease, it has no impact on their personal relationship, and it is guaranteed that no one else ever finds out. Oftentimes, queried participants respond with strong and stubborn evaluations that it is morally wrong for them to have consensual incest, even when reminded that there is no harm caused (2000, p. 9). They maintain a judgement of moral wrongness, simply because it just *is*¹.

While moral dumbfounding provides strong evidence against moral judgements stemming from a solely rational process, it does not offer evidence in favour of a specific emotion-based alternative. As such, a variety of theories exist which each aim to explain what an emotion-based morality may look like. Haidt and Joseph propose a conception of morality that is both innate and learned (2004). They argue that humans come equipped with a number of moral modules that form the basis of moral systems. Modules can be thought of as black-box mental processing units, evolved as a result of repeated exposure to situations which demand a certain type of processing throughout human history. Haidt and Joseph's proposed five moral modules include suffering, hierarchy, reciprocity, purity, and possibly, an in-group module (2004 p. 59, 63). These modules give rise to flashes of moral intuitions, from which moral judgements are made. However, these innate modules can be modified and manipulated based on cultural inputs, leading to differing intuitions based on cultural context.

I would like to note that these moral intuitions, as described, parallel strongly to a System 1 type of cognition. Dual-process theories of cognition categorize human cognition into two different modes of thinking. System 1 processing can be described as automatic, fast, and uncontrolled. This processing accounts for snap judgements that are formed from a variety of sources, including emotions, heuristics, and intuitions. While System 1 judgements happen almost immediately, the processes followed to get there can be changed over time. That is, the mechanisms for forming intuitions can be consciously molded and rationally altered. But it is entailed that at the moment of an intuition, the experiencer is unaware of the process that resulted in that flash of insight. In relation to the realm of morality, intuitions refer to an innate predisposition to experience whiffs of rightness and wrongness in the face of a moral situation.

¹ It is important to note that MD does not suggest that people cannot provide reasonable explanations behind their moral judgements after the judgement is made. However, retroactively providing reasons for a judgement does not mean that those reasons were explicitly used in its formation. To be specific: MD only entails the phenomenon where a judger clings to a moral judgement even when they are unable to provide reasons behind it.

For the remainder of this paper, when “intuition” is used, I am referring to a judgement formed from the entire umbrella of sources which feed into this System 1 type of processing.

On the other hand, System 2 processing refers to a slow, intentional, and controlled cognitive process. When you are consciously reasoning through a problem, or working backwards from some goal to determine the best course of action, or using reason to override an unsavoury gut-instinct, you are employing a System 2 type of processing. The conclusions reached through System 2 processing can be influenced and altered by factors such as education and reflection. While conclusions reached through System 1 processing are more difficult to alter, doing so is still possible — for example, through the process of repeatedly altering one’s habits. System 2 processing is also capable of influencing System 1 conclusions through either inhibiting or drawing increased attention to certain subsystems. In other words, System 2 processing can influence System 1 processing: intuitions can be shaped through knowledge to be perceptive to different contexts².

A second theoretical perspective on moral judgement is the Linguistic Analogy (LA), which draws a comparison between Chomsky’s theory of linguistic cognition and moral cognition (Dwyer et al., 2009). Similarly to the proposition that our brains contain some innate linguistic grammar that allows humans to acquire and use language, LA suggests that our minds come equipped with an innate *moral* grammar that allows humans to acquire and use morality. Where we can form an infinite number of sentences, we also seem to be able to navigate our way through an infinite number of moral scenarios. Beyond this, just as we are able to make seemingly immediate judgements on the acceptability of any given sentence — in other words, given a sentence, we are able to judge how permissible it would be to use that sentence regardless of its grammaticality — we are also able to make acceptability judgements on a seemingly infinite number of moral scenarios (take, for example, the sheer number of trolley problem derivations we are able to quickly form judgements on). Proponents of LA argue that

² The dual process theory of *moral judgement* is a sub-domain of dual process theories of cognition with its own extensive literature. However, in this paper, when I use the term “dual process *intuitionism*” I am not referencing the dual process theory of moral judgement as initially proposed by Joshua Greene (Greene et al., 2001). There is overlap in this discussion, but for the scope of this paper, dual process intuitionism refers explicitly to Haidt’s conception of intuitionism, with my added point that it is worthwhile to think of Haidt’s intuitions as a System 1 process, situated in the broader domain of dual process theory.

I would also like to note that for the purposes of this paper I will be assuming the general picture of dual process theory. There are a number of complexities within dual process theory, and perhaps most pertinently, around just how much has been packed into conceptions of System 1 on many accounts. For example, certain conceptions of System 1 also consider it to be *associative* (Sloman, 1996), which would clearly cause some troubles in the realm of moral intuitions. As such, in the interest of exploring the complexity with which intuitions may handle MD, I will be assuming the basic picture of System 1 and System 2 processing. Specifically, that System 1 processing refers to rapid, uncontrolled processing.

we come equipped with some mental structure and innate moral grammar which allow us to navigate an unbounded number of moral situations (Dwyer et al., 2010).

In her 2009 essay, “Moral Dumbfounding and the Linguistic Analogy”, Susan Dwyer argues that in the wake of the revelations spurred by the phenomenon of moral dumbfounding, the linguistic analogy has an advantage over other emotion-based theories for explaining the moral structure of the mind in that it encompasses the “complexity” required of cognition related to moral reasoning. More specifically, she argues that moral judgement-makers must be able to “distinguish between the causes and consequences of actions” and to account for “the intentions of agents”, which emotions and rapid cognition are not able to do (p. 285). Dwyer formalizes this proposed requirement of moral judgements as such: that a judgement “must accommodate subjects’ grasp of the structure of the scenes they evaluate” (p. 294). While moral dumbfounding calls into question theories of moral psychology whereby moral judgements are formed based on consciously accessible reasoning, Dwyer argues, the Linguistic Analogy preserves complexity by allowing for structure in our moral processes, and constraints on this mental structure such that moral acceptability judgements can be made.

It is here where my aim for this paper begins. I do not agree that intuition-based explanations for moral judgements are incapable of harnessing the situational awareness or cultural context required for moral judgement making. I think this is especially pertinent when we conceptualize these intuitions as a result of System 1 processes, which can work in tandem with System 2 processes. Dwyer suggests that LA uniquely allows us to inquire into how creatures like ourselves are able to “make moral judgements at all” (2009, p. 294), but by making an analogy of my own to the sheer number of inputs which allow us to make an unbounded number of complex judgements in safety scenarios, I hope that the reader can see that intuitionist approaches to morality allow us to accomplish this same thing. In this way, I do not think that the Linguistic Analogy has quite the advantage over other emotion-based explanations for moral judgement-making as Dwyer would hope.

PART 2: Gut Feelings and Complex Intuitions

Scenario 1: Taylor has a particular way that she walks home every day. She has never had an issue taking this route and has always made it home safely. One day, Taylor has a hesitant gut feeling about one street on the route, and decides to take another way home. If she was asked to explain why she thought it best to avoid that street in this instance, she may be unable to provide a specific reason. But that’s not to say that Taylor is acting on impulse, or solely because of anxiety.

Perhaps one of the streetlights had burnt out, and the street was more dimly lit than she was comfortable with. Perhaps she had heard a news report earlier in the day about an unsavoury character being loose in the area, and while she did not remember it at the time, it was enough to present her with an uncomfortable feeling. Or perhaps she caught a figure moving in the street out of the corner of her eye, and while she did not consciously recognize it, she was able to intuit that it would be best to avoid it.

Scenario 2: Sophia is planning on getting a ride to the store with her friend Chris. Chris has driven them both many times without incident, and has always seemed to be a reliable driver. However, this time, when Sophia gets in their car, something seems off. She decides she'd rather take the bus, and when Chris asks why he becomes irritated, because Sophia is unable to provide him with a reasonable explanation as to why she'd prefer not to drive with him that day. While she may not be consciously aware of it, maybe Chris was showing signs of not getting enough sleep the night prior, or maybe he seemed more on edge than usual. Or maybe, in the past he had made a flippant remark about driving after having a few beers. While this is not top of mind for Sophia in the moment of her decision not to drive with him, perhaps that information has made her more hyper-aware of Chris's condition when driving.

In both cases, while the subjects may not have had access to the reasoning behind their safety judgements, we would be hard pressed to say that they were forming these judgements solely on mood or emotions. It wasn't isolated anxiety that caused Taylor to opt for a different route home. Although on the surface this scenario seemed identical to her previous commutes, there was enough of a contextual difference — either in that immediate moment or in new information Taylor had learned since the last time she commuted — that she experienced a safety intuition which motivated her to take a different route. A similar process occurred in Sophia's case. Taylor and Sophia's gut feelings were able to incorporate context-specific information, which contributed to their safety judgements.

In the same breath, we also do not seem to require that humans have some sort of innate safety grammar, which allows us to make an unbounded number of safety judgements in both social and non-social situations. The number of perceptual inputs which provide information in safety scenarios, combined with a variety of cognitive and physical systems —

some specifically related to safety, and some not — seem sufficient for us to navigate unique and complex situations. For example, in application gut instincts are leveraged as a source of information for midwives in cases of suspected domestic violence among patients (Baird et al., 2020). This is not to say that gut instincts are infallible, or should always be used over a rational process of thinking. I use these examples to belabour the point that intuitions are capable of being complex and context-dependent enough that they can be included as an important aspect of decision-making.

Because safety intuitions are capable of this level of complexity, I do not see why moral intuitions are incapable of doing the same. Just as our safety intuitions may be founded on underlying reasons or principles (i.e. Chris's lackadaisical attitude towards inebriated driving), so can our moral intuitions. For example, Haidt and Joseph's proposed purity module may underly the intuition that consensual incest is wrong, even when it does not bring about any harm. The judger may not be able to articulate this when asked, but this doesn't mean that it did not contribute to their intuition that it is morally wrong. And just as our safety intuitions can take into account situational context, so too can our moral intuitions. Take, for example, the stereotype of a date who is rude to the waitstaff. While being rude to another person may not be sufficient for the actor to be written off as a morally frowned upon individual, there is something specific to the scenario of a diner impolitely interacting with the server waiting on them that seems to elicit a pang of moral wrongness, even if the observer is unable to articulate why this instance of rudeness is worse than any other rudeness.

It is here where I think we can also begin to see the advantage of a kind of dual process intuitionism. For example, our hierarchical module may suggest an intuition whereby wait staff should be deferential to the people they are waiting on, and exhibit some level of respect that is worthy of reproach if not adhered to. But our cultural context, which maintains that all people regardless of profession should be treated with respect, can alter this intuition. Our System 2 reasoning can de-value the hierarchical module in this context within our System 1 intuition-making processes, through education from your friends or teachers or colleagues, that this behaviour is an indicator of poor character. As such, when we see someone act rudely towards a server, we have a predisposed inkling that what they are doing is a moral wrong, even if we do not consciously have access to this process at the time.

Through this exploration into the complexity of intuitions generally, Dwyer's suggestion that the linguistic analogy is advantageous over intuitionism in its ability to take into account causes, consequences, and intentions in moral scenarios seems to be diminished. In fact, LA may have its own disadvantage when compared to intuitionism, in that it is unable to explain the

cultural variance in attitudes towards moral scenarios that are said to reveal information about underlying moral cognitive structures. LA's applicability is confined by the existence (or lack thereof) of universal acceptability judgements, and I argue that dual-process intuitionism does not face this same issue.

PART 3: The Linguistic Analogy and Cultural Variance

By comparing moral acceptability scenarios — scenarios eliciting *permissibility* judgements, not necessarily *rightness* or *wrongness* judgements — to linguistic acceptability scenarios, Dwyer bears the requirement that moral acceptability judgements should be largely universalizable. That is, just as we use universalized judgements on linguistic acceptability to learn more about the cognitive systems that allow us to acquire language, we must also have universalized judgements on moral acceptability scenarios to signal that humans share moral cognitive systems. The issue, here, is that the moral acceptability scenarios Dwyer presents don't actually result in universalized judgements.

Take Dwyer's two presented trolley problem scenarios. In the first, you can save five people tied to a track by pulling a lever which diverts the trolley to another track containing one person; in the second, you can save the five people on the track by pushing a large person from a bridge onto the track and stopping the trolley (2009, p. 292). Dwyer frames these problems as moral acceptability scenarios, à la linguistic acceptability scenarios. Through the lens of the LA, a majority of respondents reporting that scenario one is permissible (or acceptable) and scenario two is impermissible (or unacceptable) provides credence that there is some innate moral structure advising this acceptability judgement.

One thing Dwyer does not take into account, however, is the cultural variance in judgements on these examples (Ahlenius & Tännsjö, 2012; Gold et al., 2023). If these trolley problems are actually acceptability scenarios, as Dwyer purports, then we should expect similar acceptability judgements within similar cultures, and at best, across cultures. But we do not. In Ahlenius and Tännsjö's study, using these same scenarios, 81% of American respondents said that pulling the lever in scenario one was permissible, while only 52% of Chinese respondents said that it was permissible (2012, p.197). For scenario two, 39% of American respondents said pushing the large person was permissible, while only 22% of Chinese respondents said it was permissible. If we say that acceptability judgements should at the minimum result in similar assessments *within* a culture, Chinese responses to scenario one being seemingly up to chance

suggests that judgements regarding these acceptability scenarios may not be so concrete after all³.

This cultural variance can be explained through dual-process intuitionism. There may be an underlying module which explains general trends towards permissibility judgements in these scenarios (perhaps the suffering module). However, the way this module manifests in intuitions can be shaped by cultural attitudes towards, for example, responsibility to act, utilitarian-oriented values, and fate (Gold et al., 2023). For instance, if you grew up in a culture that placed a lot of value in the role of fate, your suffering module could be altered so that your intuitions suggest that you not intervene in events that could be viewed as fatalistic. While LA may be constrained by having to expect common responses to acceptability scenarios — at least within a culture that shares a similar moral outlook — dual-process intuitionism does not share this same problem.

PART 4: Conclusion

While moral dumbfounding has played its part in re-invigorating the discussion surrounding emotive involvement in moral decision-making, I am hesitant to write off intuitions as incapable of providing the contextual sensitivity needed to form moral judgements. Our intuitions whisper to us when something isn't quite right, or when we should be hyper-aware of an individual, or when we should stick up for what we feel is true — despite a lack of accessible explanation. A dual-process intuitionism, which builds on top of Haidt and Joseph's proposed moral modules, is able to advise context-sensitive and complex moral judgements through the existence of constrained base intuitions which can be refined and altered through informational input and cultural intervention. By re-framing emotive involvement in moral judgement-making, not as impulsive or crude, but as sensitive and grounded — while still requiring regulation and refinement so as not to become perverted — I feel as though we are able to have a much more insightful look into how our heads, and hearts, play a role in our morality.

³ While research on trolley problem responses is widespread (Cushman et al., 2006; Hauser et al., 2007), it has largely been performed on western, educated, industrialized, rich, and democratic (WEIRD) populations. Gold et al. argue that Chinese culture in particular is worthy of further research due to Chinese populations' differing responses to what were previously thought to be universal responses to trolley problem scenarios (2023).

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